

ANSHUL AGARWAL

Bengaluru, India | +91-7706834062 | anshulanshul54@gmail.com
linkedin.com/in/anshul-eng | github.com/agarwaltech

SUMMARY

AI & ML engineer with DevOps and QA automation experience, focused on AI-driven build and release workflows. Builds CI/CD pipelines using GitHub Actions and Docker, deploys and monitors containerized services on AWS and Kubernetes, and integrates automated testing with Pytest, Selenium, and Postman as enforced quality gates. Applies LLMs to DevOps problems including automated build-failure triage, log analysis, and test generation. Strong Linux, Python, and Bash scripting foundation.

TECHNICAL SKILLS

CI/CD & Automation: GitHub Actions, Jenkins, Git, GitHub, branching and pull-request workflows, build and release automation, artifact management

Containers & Orchestration: Docker, multi-stage builds, Dockerfile optimization, Docker Compose, Docker Hub, Amazon ECR, Kubernetes (Deployments, Services, ConfigMaps, HPA), Helm

Infrastructure as Code: Terraform (modules, remote state), Ansible, YAML

Cloud Platforms: AWS (EC2, S3, IAM, ECR, VPC, CloudWatch, Lambda), Azure (basics), cloud deployment, IAM and secrets management

Monitoring & Observability: Prometheus, Grafana, CloudWatch, alerting rules, structured and centralized logging

QA & Test Automation: Pytest, Selenium, Postman and Newman API testing, unit, integration and regression testing, smoke and sanity testing, STLC, test case design, defect tracking and bug lifecycle (JIRA), coverage reporting, Trivy image scanning

Generative AI & Agentic Systems: LLMs, Prompt Engineering, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Embeddings, Vector Databases (FAISS, Chroma), Tool Calling, Multi-Agent Systems, FastAPI model serving

AI & Machine Learning: Supervised Learning, Classification, Feature Engineering, Model Evaluation, Natural Language Processing (NLP), Scikit-learn, Pandas, NumPy

Programming & Scripting: Python, Bash, SQL, JavaScript, Linux (Ubuntu), shell scripting, cron

Core Concepts: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Mind Matrix

Jan. 2026 - June 2026

GenAI Intern

Remote

- Containerized and deployed FastAPI-based LLM services using Docker, standardizing environments across development and production and eliminating local dependency conflicts.
- Automated build and deployment steps through Git-based workflows and Python and Bash scripts, cutting manual release effort across 3 shipped GenAI features.
- Added structured request logging, error handling, and retry logic to API services, reducing time spent diagnosing production failures.
- Built RAG and vector-retrieval pipelines with embeddings and structured outputs for production AI use cases.

PROJECTS

AI-Powered CI/CD Failure Triage Bot | [GitHub Actions](#), [Python](#), [Docker](#), [Groq LLaMA 3.1](#) | [\[Code\]](#)

- Built a GitHub Actions workflow that captures logs from failed builds and sends them to an LLM that classifies root cause (dependency, test, lint, or infrastructure) and posts a plain-English summary with a suggested fix as an automated pull-request comment.
- Removed manual log-scrolling from the debug loop by automatically surfacing the failing step, exact error line, and remediation hint on every failed run.
- Packaged the analyzer as a reusable Docker-based composite action with configurable model, severity thresholds, and Slack webhook notifications.

End-to-End CI/CD Pipeline for a Containerized FastAPI Service | [GitHub Actions](#), [Docker](#), [AWS EC2](#), [Pytest](#), [Trivy](#) | [\[Code\]](#)

- Designed a multi-stage pipeline running lint (flake8), unit and integration tests (Pytest), multi-stage Docker build, Trivy vulnerability scan, image push to Amazon ECR, and automated deployment to AWS EC2.
- Reduced final image size by roughly 60% using multi-stage builds and slim base images, and gated merges on a test-coverage threshold so failing builds never reached the main branch.
- Managed credentials with GitHub Encrypted Secrets and scoped IAM roles, and implemented health-check-based rollback to the last working image on a failed deployment.

Automated QA & Regression Suite Integrated into CI | [Pytest](#), [Selenium](#), [Postman/Newman](#), [GitHub Actions](#) | [\[Code\]](#)

- Built an automated regression suite combining Pytest unit tests, Selenium browser tests, and Newman API collections, triggered on every pull request through GitHub Actions.
- Designed smoke, sanity, and full regression test layers with reusable fixtures and data-driven test cases covering positive, negative, and boundary conditions.

- Published HTML test reports and coverage artifacts per run and configured the workflow to block merges on any failing test, creating an enforced quality gate, with nightly scheduled runs against staging and automated failure notifications.

Terraform + Kubernetes Deployment with Prometheus & Grafana Monitoring | *Terraform, Kubernetes, Helm, Prometheus, Grafana, AWS* | [\[Code\]](#)

- Provisioned reproducible AWS infrastructure (VPC, subnets, security groups, compute, S3 remote state) using modular Terraform, enabling full teardown and rebuild of the environment with a single command.
- Deployed a containerized application to Kubernetes using Deployments, Services, ConfigMaps, and a Horizontal Pod Autoscaler, and verified rolling updates, self-healing, and zero-downtime releases.
- Instrumented the cluster with Prometheus and Grafana dashboards tracking CPU, memory, pod restarts, and request latency, with alert rules for crash-looping pods.

EDUCATION

Sir M. Visvesvaraya Institute of Technology, Bengaluru
Bachelor of Engineering in Artificial Intelligence and Machine Learning

Dec. 2022 - June 2026
CGPA: 8.5

ACHIEVEMENTS

- Published a research paper based on the Toxibot project.
- Led the Under25 campus team for events and student outreach.
- Earned NCC B and C Certificates, demonstrating leadership, discipline, and national service commitment.

CERTIFICATIONS

- AWS Certified Cloud Practitioner - Amazon Web Services.